Md Zaman

Software Development Engineer in Test (SDET)

Contact

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- GitHub: https://github.com/gits5213
- Portfolio: https://mdszaman.com
- Training Website: https://gitsics.com

Work Status

- US Citizen

Education

- National University of Bangladesh (Govt. Titumir College)
  - Masters in Chemistry

Licenses & Certifications

AWS solution architect associate

ASTQB-ISTQB®

- Advanced Test Automation Engineer (CTAL-TAE)
- Advanced Test Manager (CTAL-TM)
- AI Testing (CT-AI)

Professional Scrum Master I - Scrum.org

Scaled Agile, Inc.

- Certified SAFe® 6 Practitioner
- Certified SAFe® 4 Agilist
- Certified SAFe® DevOps (6.0)

CompTIA Security+

- DHS Trusted Tester Program certification

Tricentis Tosca

- Test Architect for Tricentis Tosca (AS1, AS2, API Test, TDS1)

ACCELQ Automation Engineer (May 2025)

Volunteering

- Auxiliary NYPD (New York City Police Department) - Sep 2019 to Present.

Honors & Awards

- Spot Award – RTX - Raytheon Technologies (YOU'RE BEING RECOGNIZED FOR: INNOVATION- Mar 2020)

Expertise in Tech Stack

Programing Language:

- Python, Java, Javascript, Typescript (NodeJS) & C#.Net

Automation API & Tools:

Automation Testing: Selenium, Gauge, CypressIO, Playwright, SpecFlow, RestSharp, CodedUI, WebDriverIO, Protractor, Robot Framework, TestCafe, Xcode, Appium, XCUITest, NightWatch, RestAssured, Cucumber, Gherkin, SuperTest, Katalon Studio, SauceLabe, BrowserStack, LamdaTest, Frisby, PyTest, ReadyAPI, Jupiter notebook, Maven, Gradle, npm, Git, GitHub, Bitbucket, Eclipse, Intellij, SVN, Unix/Linux, CircleCI, Jenkins, AWS, S3, IAM, EC2, Amazon Connect, DynamoDB, Cloud Watch, Azure, EventHub(Event-driven microservices), Kafka, GraphQL, Axe, Pa11y, PowerBI, Tableau, SOAPUI & REST API, TMW, Tosca, Generative AI (ChatGPT), Prompt Engineering, AccelQ, Subject7m, Cyara, Cyara Botium (LLM Chatbot Testing), MCP Claude, Gemini, GPT-40, CoPilot, Programming languages, PowerShell, Azure, QA/test Automation, Performance Testing.

Manual & Process:

Manual Testing & DevOps Practices, Agile-Scrum, Azure DevOps (ADO), JIRA, Xray, ALM, Rally, Zephyr, Adaptavist, SQL, Oracle, Swagger, Confluence, POSTMAN, Splunk, New Relic, SonarQube & CheckMarx, BigData, Figma, JSON, HTML, Immuta, AWS MySQL, Microsoft SQL Server, PostgreSQL, Snowflake, DBeaver, PGAdmin, JMeter & Locust (Load & Stress test), Docker, WebRTC, FIX, REST, Lighthouse, WAVE, JAWS, NVDA, PDF, ANDI, Wireshark, Ampliance, ComerceTool, Bug Tracking, Quality Assurance, SDLC, Defect Tracking Tools, Analytical Thinking, Software QA Methodologies, Scripting, Software Development Process, Client Handling, Object-oriented Programming, Performance Testing, Security Testing.

Summary

Tech Lead Automation Engineer & Senior Test Automation Engineer with 15+ years of experience in software and hardware testing, specializing in UI Testing, Mobile Testing, Middleware Testing, API testing (SOAP/REST) and automation within Agile DevSecOps environments. Proficient in tools like SoapUI, REST Assured, Selenium, Cypress, Playwright and JMeter for automation Testing and performance testing. Skilled in CI/CD integration using GitLab and cloud platforms (AWS, Azure). Strong background in SQL for data validation and a proven ability to deliver high-quality, secure software through effective collaboration and leadership.

Industry Experience:

Analytics, Cyber Security, Information Services, Information Technology, Project Management, Security, Banking, Finance, Trading, Ecommerce, Insurance, Health, Digital Media, Telecommunication, Blockchain, Aviation & Defense, Education, Logistics, CRM, CMS, Salesforce, Mainframe, Biocommerce, MuleSoft, Magento & Windows, macOS, Unix/Linux, Kubernetes, Salesforce, SAP.

Tech Lead/Architect, QA Engineer (SDET) - Global iTech Solutions **Sept 2024 - Present**

Lead QA and **automation strategy for enterprise** clients (e.g., HD Supply-Internal Inventory Management Microservice) and Centene (Contract Center), collaborating with business and engineering teams to enhance end-to-end testing processes, implement quality standards, and drive innovation in test automation.

- Leading end-to-end quality engineering for clients (e.g., HD Supply), collaborating with business and engineering teams to deliver precise, high-quality products.
- Lead end-to-end **quality engineering** for enterprise clients, partnering with business and engineering teams to deliver reliable, scalable, and high-quality software solutions.
- Drive **QA strategy and test modernization** through automation standards, **508 accessibility compliance**, Agile **3-Amigos** practices, and real-time QA metrics and dashboards.
- Design, build, and maintain automation frameworks across **UI, API, middleware, accessibility, and mobile testing** using **Playwright, Cypress, Appium, ReadyAPI, GraphQL**, and AI-powered testing tools.
- Leverage experience with existing **Tricentis Tosca** suites to support migration to **Playwright with TypeScript**, improving automation speed, stability, maintainability, and scalability.
- Lead automation transformation by migrating legacy test coverage to **Playwright with TypeScript**, enabling faster execution, stronger regression coverage, and improved **in-sprint automation**.
- Lead automation transformation by migrating legacy test coverage from **Katalon Studio to Playwright with TypeScript**, enabling faster execution, stronger regression coverage, and improved in-sprint automation; continuously worked in **Katalon Studio** during the transition and now continue driving automation development in **Playwright with TypeScript**.
- Managing & Preforming **SAP Automation** and end-to-end testing across **SAP GUI, SAP Fiori, SAP ERP & Apriso Web App, SAP EECC, and S/4HANA,** validating critical business process across FICO, MM, SD, and HCM Modules leveraging **Tricentis Suite (Tosca, qTest, LiveCompare, NeoLoad, Data Integrity)**
- Design, Develop and scale **enterprise regression automation suites** for complex applications, applying expertise in **Tosca** and **Playwright** to support sustainable test strategy and smooth framework migration.
- Implement **API automation frameworks from scratch** using **Postman** and **Newman**, integrating them with **GitHub Actions CI/CD pipelines** for nightly execution.
- Develop **mobile automation frameworks from scratch** using **Appium** for **iOS and Android** with **WebdriverIO**, integrating with **LambdaTest** for emulator and real-device coverage and **GitHub Actions** for CI execution.
- Create and execute **performance testing solutions** for **UI, API, and database load testing** using **BlazeMeter** and **JMeter**, with continuous tuning based on sprint and project needs.
- Design and maintain **CI/CD-integrated automation solutions** across **GitHub Actions** and **AWS services** including **S3, DynamoDB, CloudWatch, Amazon Connect, and Genesys**.
- Support **performance, security, and GraphQL modernization initiatives**, including optimization of system behavior under load and migration of critical endpoints.
- Conduct **proofs of concept** across **Robot Framework**, **Katalon, AccelQ, and Tosca** to evaluate tools, improve automation efficiency, and align testing solutions with business goals.
- Hands-on expertise in **UI, API, Middleware, Accessibility, and Mobile Automation** using **CypressIO, Playwright, Selenium, Appium, TestCafe, ReadyAPI, GraphQL**, and **AI-powered** testing (**Claude Code, ChatGPT, CodeGPT, Copilot, Gemini**).
- Mentor QA engineers and cross-functional teams in **automation best practices, accessibility, Performance Testing, CI/CD integration, and AI-driven testing**, strengthening quality culture and delivery maturity.

Tech Lead/Architect Engineer (SDET) – American Airlines (via Insight) **Sept 2023 – Sept 2024**

Supported internal portal for merger management microservices across M&A, capital markets, trading, and liquidity/payment domains.

- Led QA efforts for **Payment Modernization** program, testing payment authorization, capture, refund, cancel, 3DS, and reconciliation via **Adyen gateway** with PostgreSQL backend.
- Defined and implemented **manual, automation, accessibility, and performance testing strategies**, improving test maturity across teams.
- Migrated frameworks from **NightwatchJS/Protractor → Cypress → Playwright**; transitioned REST API endpoints to **GraphQL**.
- Designed and built **cross-platform test automation frameworks** (UI, API, mobile, microservices) using Selenium, Playwright, Cypress, Cucumber, Python, Java, C#, JS, SoapUI, ReadyAPI, Postman.
- Set up **CI/CD pipelines** with AWS, ADO, Jenkins, SonarQube; leveraged Docker, BrowserStack, SauceLabs, LamdaTest for scalable test execution.
- Mentored and coached junior engineers on **OOP principles, automation best practices, and tools** (Selenium, Cypress, AccelQ, WebDriverIO, Python, C#).
- Drove **security & accessibility testing** (SonarQube, AXE, WAVE, JAWS, NVDA, PA11Y) and **performance testing** with JMeter.
- Validated **ETL/data pipelines** using Python & PySpark ensuring data integrity in transformations/migrations.

Key Achievements:

- Delivered robust automation frameworks across **Web, API, Mobile & Middleware** with reusable architecture, reducing test cycle time.

- Standardized test practices and ADO scorecards, enabling better visibility into QA quality metrics.
- Proactively reduced technical debt and complexity while improving test environment reliability.

Sr. Mgr. Software Dev - Raytheon Technologies (RTX) **Oct 2018 - Sept 2023**

Supported digital accelerators across aviation, defense, and space programs, delivering intelligent QA solutions for enterprise applications.

- Led QA initiatives within Enterprise Digital Accelerators (EDX), establishing manual, automation, accessibility, and performance testing best practices.
- Drove migration of automation frameworks (**CypressIO → Playwright**), improving efficiency and reliability.
- Drove migration of automation frameworks (**Robot Framework → Playwright**), improving efficiency and reliability.
- Designed and implemented boilerplate architectures for **UI, API, and Middleware (event-driven microservices) testing frameworks** using Java, Python, C#, JavaScript, and cloud services (AWS, ADO, Jenkins, SauceLabs, BrowserStack, LamdaTest).
- Built automation frameworks from scratch for multiple mission-critical projects (Sensitech STM, Reliability Prism, FPO, Rotor Design System, XetaCloud, Digital Twin, SAP Migration).
- Performed ETL and database validation with Python & PySpark to ensure data integrity; conducted accessibility 508 compliance testing using AXE, Pa11y, JAWS, NVDA.
- Delivered QA Metrics Dashboards and standardized processes across EDX to streamline reporting and governance.
- Mentored engineers in OOP, automation frameworks, and containerization (Docker), driving knowledge sharing and consistency.
- Hands-on contributor in test automation, CI/CD integration, version control (GitHub, Bitbucket, GitLab, ADO, SVN), and Unix/Linux scripting.

Technical Lead & Hands-on on a project ‣ Accomplishment:

- Sensitech STM Project (Hybrid Application) - Reliability (Prism) Project (Web Application) - FPO (Flight Optimization Profile) - Rotor Design System (CSS Framework) - XetaCloud (Portal - DataPlatform) - Digital Twin (Portal), SAP Migration & QA Metrics Dashboard.
- Automated from the scratch to advance level framework for the above microservice application leveraging Protractor, CypressIO, Robot Framework, Playwright, AXE, Pa11y, ReadyAPI, Postman, Javascript, Java, TestNG, Maven, Python, PyTest, JMeter, Locust, Appium (XCUITest) and leveraging cloud hosted service SauceLabe, BrowserStack & LamdaTest, AWS, AzureDevOps (ADO), NightwatchJS, Accessibility 508 compliance testing, SoapUI & Restful API.
- Across EDX - Set up SQA (Manual & Automation) Standards & Best Practices.
- Across EDX - Set up UI, Middleware & API's Multi Automation Framework boilerplate architecture.

Key Achievements:

- Established **enterprise-wide QA standards and multi-layered automation frameworks**, enabling scalability across EDX.
- Improved testing efficiency through automation coverage of hybrid/web apps, APIs, and data platforms.
- Enhanced accessibility compliance and integrated performance/load testing into CI/CD pipelines.

Lead Software Engineer in Test - Bank of America Merrill Lynch (via Mitchell Martin) **May 2018 - Oct 2018**

Supported internal portal merger management microservice application for equity, lending, trading, risk management, and payments services.

- Partnered with business, engineering, and operations teams to improve end-to-end testing processes, tools, and methodologies.
- Designed manual, automation, accessibility, and performance testing strategies leveraging industry best practices.
- Led test automation and CI/CD integration using GitHub, Bitbucket, GitLab, ADO, AWS, Jenkins, SonarQube, and SauceLabs.
- Hands-on with Unix/Linux commands and cloud-based test environments.

Key Achievements:

- Enhanced agile *3 Amigos* process and documentation, streamlining collaboration.
- Built automation from scratch for Web, iOS, and Android microservices using Cucumber, Java, Selenium, XCUITest, Android Studio, TestNG, Maven, AXE, SoapUI, GraphQL, and Allure Reports.
- Conducted ETL and database validation with Python & PySpark ensuring data accuracy and integrity.
- Delivered manual and automated accessibility testing on Web/PDF using ANDI, JAWS, NVDA, AXE, WAVE, and PA11Y.
- Automated internal web apps with Cypress (POM, data-driven) integrated into CircleCI, Docker, and AWS (S3).

Sr. DevOps/Automation Engineer - Hearst Magazines **Oct 2017 - Apr 2018**

Supported digital strategy for 30+ magazine brand websites across fashion, beauty, lifestyle, automotive, food, and family domains.

- Partnered with business, engineering, and operations teams to refine end-to-end testing processes, tools, and methodologies.
- Designed manual, automation, accessibility, and performance testing strategies using industry best practices.

- Maintained source control and daily automation tasks with GitHub, Bitbucket, GitLab, ADO, SVN, and Unix/Linux.

Key Achievements:

- Optimized existing NightwatchJS & Protractor frameworks and migrated to CypressIO with Page Object Model, data-driven, and keyword-driven concepts.
- Automated API (REST & GraphQL) testing using RestAssured & SuperTest, integrated with AWS, ADO, Jenkins, SonarQube CI/CD pipelines.
- Conducted ETL and database validation with Python & PySpark to ensure accuracy and reliability in data transformations.
- Delivered manual and automated accessibility testing on Web/PDF using ANDI, JAWS, NVDA, AXE, WAVE, and PA11Y.
- Utilized Docker for consistent test environments, dependency management, and CI/CD efficiency.
- Conducted manual API testing with Postman, Swagger, and ReadyAPI before automating workflows.

Lead, Software Engineer in Test – JCDecaux Mar 2017 - Oct 2017

Supported **Billing Management (BillIT)** and **Posting Management (PostIT)** microservice applications.

- Collaborated with business, engineering, and operations teams to optimize end-to-end testing processes and methodologies.
- Developed manual, automation, accessibility, and performance testing strategies aligned with industry standards.
- Managed version control and daily automation tasks using GitHub, Bitbucket, GitLab, ADO, SVN, and Unix/Linux.

Key Achievements:

- Built automation frameworks from scratch using NightwatchJS & Protractor, integrated with CI/CD pipelines.
- Automated RESTful & GraphQL API endpoints leveraging RestAssured & SuperTest; integrated with AWS, ADO, Jenkins, and SonarQube.
- Delivered manual and automated accessibility testing for Web/PDF with JAWS, NVDA, ANDI, AXE, WAVE, and PA11Y.
- Implemented Docker for reliable test environments, dependency management, and CI/CD efficiency.
- Conducted API testing with Postman, Swagger, and ReadyAPI before automating workflows.

Lead, Software Engineer in Test - Cloud9 Technologies Jul 2016 - Mar 2017

Contributed to Cloud9 trading platform (desktop & web) microservice applications.

- Worked closely with the business groups, engineering organizations, software development teams, and operations teams to continually evaluate and improve the end-to-end software development in test processes, tools, and testing methodology used in a project.
- Collaborated with business, engineering, and operations teams to refine end-to-end testing processes, tools, and methodologies.
- Designed manual, automation, accessibility, and performance testing strategies using industry best practices.
- Maintained daily automation workflows and source control using GitHub, Bitbucket, GitLab, ADO, SVN, and Unix/Linux.

Key Achievements:

- Automated **desktop trading app** from scratch using C#.NET & NUnit; integrated with CI/CD pipelines.
- Automated **web applications** (user management & support) using NightwatchJS, Protractor, SuperTest, with AWS, ADO, Jenkins, SonarQube CI/CD integration.
- Delivered manual & automated **accessibility testing** for web/PDF using ANDI, JAWS, NVDA, AXE, WAVE, and PA11Y.
- Utilized **Docker** for consistent test environments, dependency management, and CI/CD optimization.
- Conducted API testing with **Postman, Swagger, ReadyAPI**, automated with C#.NET (NUnit) & SuperTest.

Lead Automation Engineer ASE6 - Wells Fargo (via TEKsystems) Nov 2015 - Jul 2016

Supported Customer Information View (CIV) and Contact Center microservice applications.

- Partnered with business, engineering, and operations teams to improve end-to-end testing processes, tools, and methodologies.
- Created manual, automation, accessibility, and performance testing strategies leveraging industry best practices.
- Maintained source control and automation workflows with ADO, Jenkins, Git, and cloud-based integrations.

Key Achievements:

- Enhanced Selenium, Cucumber, Java, TestNG framework by applying KISS/DRY coding standards and QA best practices.
- Automated contact center master call flow with Empirix & Cyara across multiple IDEs (VS Code, PyCharm, Eclipse, IntelliJ).
- Developed a new API automation framework using RestAssured (from scratch) and integrated with AWS, ADO, Jenkins, SonarQube, AccelQ, and SauceLabs.
- Conducted manual and automated RESTful API testing with Postman, Swagger, ReadyAPI, SoapUI, and SuperTest.
- Performed and automated accessibility testing on Web/PDF using ANDI, JAWS, NVDA, AXE, WAVE, and PA11Y.

SDETS ITS3 -DSHS Children administration **Jan 2015 - Nov 2015**

Supported FamLink and other CATS applications, the state's SACWIS case management system used by public and private caseworkers to maintain official child and family service records.

- Collaborated with business, engineering, and operations teams to improve testing processes, tools, and methodologies.
- Created manual, automation, accessibility, and performance testing strategies aligned with industry standards.
- Performed version control and automation tasks using GitHub, Bitbucket, GitLab, ADO, SVN, and Unix/Linux.

Key Achievements:

- Automated end-to-end case management workflows for Web and Desktop apps using Java, C#.NET, Selenium, Cucumber, CodedUI, XCUITest, Android Studio, JMeter, and SQL.
- Integrated automation with CI/CD pipelines and cloud environments via ADO, Jenkins, and SonarQube.
- Conducted API testing with Postman, Swagger, and ReadyAPI; automated with C# NUnit for maintainable coverage.
- Performed manual and automated accessibility testing for Web/PDF using JAWS, NVDA, ANDI, AXE, WAVE, and PA11Y.

Sr. Test Engineer - AT&T (via Pyramid Consulting & Tech Mahindra) **Jun 2014 - Dec 2014**

Supported Sales, Service & Support web domain, driving improvements in test processes and quality assurance practices.

- Collaborated with business, engineering, and operations teams to optimize testing methodology, tools, and processes.
- Designed manual, automation, accessibility, and performance testing strategies aligned with industry best practices.
- Maintained source control and daily automation tasks using Git, GitHub, and Unix/Linux.

Technical Lead & Hands-on contributor on a project ‣ Accomplishment:

- Built a performance testing framework using Keynote & JMeter, monitored with Dynatrace.
- Designed, developed, and automated test cases with Java, Cucumber, Selenium (API), TestNG, Maven across multiple IDEs.
- Performed manual and automated accessibility testing on Web/PDF with ANDI, JAWS, NVDA, AXE, WAVE, and PA11Y.
- Automated RESTful API testing with Java RestAssured after manual validation via Postman, Swagger, and ReadyAPI.
- Delivered detailed test design, estimations, and peer code reviews to improve test coverage and reliability.

IT Specialist/BSSO - BRAC Bank **Feb 2008 - Jun 2011**

I worked with financial domains and web-based applications. I was responsible for Manual and Automation tests both but I especially, Designed, Developed and Maintain the automation code. Analyzed the software requirements, wrote functional test cases, created test data, and performed functional tests, smoke tests, accessibility test, regression tests, and UAT tests to the functionality of the application and modules. I developed SQL queries to perform backend testing of the database. Building test automation suites for the application using Selenium (API), Hands-on SDLC (System Development Life Cycle) experience

Technology being used I have worked in waterfall software development methodologies (waterfall) • Application development, and implementation. Working with Continuous Regression Testing methodology • Selenium (API) Development, Eclipse • Java/JavaScript Development • Junit ((Java Framework) • data-driven framework development • Integration Testing, smoke test, Functional test, Regression test• UI Test Automation • Manual Test • Application Deployment, AXE, Pa11y, Lighthouse, Screen Readers, Web Accessibility Evaluation Tools.

- Performed manual accessibility testing on web and PDF files using assistive technologies leveraging JAWS and NVDA, along with tools leveraging ANDI, AXE, WAVE, and PA11Y.
- Automated accessibility testing using AXE and PA11Y to streamline and enhance the testing process for web applications and documents.
- Hands-on creating pull requests leveraging Git, GitHub, and Unix/Linux basic commands to accomplish daily automation tasks.